One Haut Encoded: A Three-Model Fashion Recommendation System on H&M Transaction Data

Dominic Tanzillo, Alex Oh
AIPI 540 Deep Learning
Duke University, Spring 2026

Abstract

Personalized product recommendation is a core challenge in fashion e-commerce, where catalog breadth and shifting trends make manual curation infeasible. We build and evaluate three recommendation architectures on the H&M Personalized Fashion Recommendations dataset [H&M Group, 2022]: a popularity baseline, a KNN content-based model on one-hot encoded article metadata, and a Neural Collaborative Filtering (NCF) deep learning model with user/item embeddings and optional metadata side features. From 31M transactions across 105K articles and 1.37M customers, we subsample to 23,373 customers and 2,998 articles using activity-based filtering. On a temporal held-out test set (last 14 days), the popularity baseline achieves the highest Hit Rate@12 (0.0998) but catastrophic catalog coverage (0.40%); KNN achieves the best NDCG@12 (0.0187) with near-complete coverage (99.2%) and highest novelty (11.85 bits); NCF with metadata features reaches 0.0178 NDCG@12 with 87.7% coverage. An ablation study comparing metadata-only, visual-only, and combined feature configurations shows that one-hot metadata features dominate over ResNet50 image embeddings for content-based filtering in this domain. The deployed interactive application at `https://one-haut-encoded.github.io` provides per-profile recommendations across all three models with a "Curate Your Look" feature for real-time personalization.

1 Introduction

H&M Group operates over 4,000 stores across 75 markets, serving millions of customers with a catalog of more than 100,000 articles spanning apparel, accessories, and home goods. The scale of this catalog creates a discovery problem: most customers see only a fraction of available products, and purchase behavior clusters around popular items regardless of personal fit. Recommendation systems address this by surfacing relevant items from the long tail of the catalog.

This project asks: **how do collaborative filtering, content-based, and hybrid recommendation approaches compare on fashion transaction data, and what role do visual features play relative to structured metadata?** Fashion is a domain where visual similarity—color, silhouette, texture—is central to consumer preference, making it a natural test case for image-augmented recommendation.

We frame this as an implicit feedback recommendation problem: given a customer's purchase history, generate a ranked list of K items the customer is likely to purchase next. The dataset contains no explicit ratings; purchases are the sole positive signal.

Our contributions are:

1. **Three-model comparison.** We implement and evaluate a popularity baseline, a KNN content-based recommender, and a Neural Collaborative Filtering model—spanning non-personalized,

similarity-based, and deep learning approaches on the same temporal test split.

2. **Feature ablation.** We compare metadata-only, visual-only (ResNet50), and combined feature configurations across KNN and NCF architectures, testing whether image embeddings add value beyond structured article attributes.

3. **Deployed interactive application.** A live GitHub Pages site with a FastAPI backend on Hugging Face Spaces provides real-time recommendations across all three models, including a "Curate Your Look" feature where users select items and receive KNN-based personalized suggestions.

4. **Precomputed fallback architecture.** The frontend ships with a 130 KB static JSON cache of recommendations for 10 demo users, enabling full offline functionality when the API backend is sleeping or unavailable.

2 Data Sources

Table 1: Data sources used in this study.

Source	Records	Access	License
H&M Transactions	31,788,324	Kaggle download	Kaggle ToS
H&M Articles	105,542	Kaggle download	Kaggle ToS
H&M Customers	1,371,980	Kaggle download	Kaggle ToS
H&M Article Images	105,542	Kaggle download	Kaggle ToS

The H&M Personalized Fashion Recommendations dataset [H&M Group, 2022] was released for a Kaggle competition in 2022 and contains three primary tables:

- **transactions_train.csv**: 31,788,324 purchase records with customer ID, article ID, and transaction date (2018-09-20 to 2020-09-22).
- **articles.csv**: 105,542 articles with structured metadata including product type, product group, colour group, department, index group, section, and garment group.
- **customers.csv**: 1,371,980 customer records with demographic attributes (age, club membership status, fashion news frequency).

Article images are organized in a hierarchical directory structure `images/{article_id[:3]}/{article_id}.jpg` enabling efficient lookup by article ID prefix.

No explicit ratings exist. All interactions are implicit (binary purchase signals). This is standard for fashion e-commerce, where browsing and purchase data are abundant but star ratings are rare.

3 Related Work

Recommendation systems for fashion and retail have been extensively studied, with approaches spanning collaborative filtering, content-based methods, and hybrid architectures.

Neural Collaborative Filtering. He et al. [2017] introduced NCF, replacing the inner product of matrix factorization with a multi-layer perceptron to learn non-linear user-item interactions. Their work demonstrated that deep architectures outperform traditional matrix factorization on implicit feedback datasets (MovieLens, Pinterest). Our NCF implementation follows their architecture with the addition of metadata and image side features concatenated to the embedding layer.

Sequential Recommendation. Kang and McAuley [2018] proposed SASRec, a self-attention-based sequential model that captures temporal dynamics in purchase sequences. While we do not implement sequential models, our temporal train/test split (last 14 days held out) reflects the

same design principle: recommendation systems must predict future behavior, not reconstruct past preferences. Sun et al. [2019] extended this with bidirectional self-attention (BERT4Rec).

Visual Recommendation. McAuley et al. [2015] explored visual recommendation using CNN features extracted from product images, demonstrating that visual similarity captures dimensions of preference that metadata alone cannot (e.g., style, pattern, silhouette). Our KNN visual variant and NCF image feature ablation directly test this hypothesis on the H&M catalog.

Content-Based Filtering. Lops et al. [2011] surveyed content-based approaches, establishing that metadata-driven systems excel at coverage and novelty while collaborative filtering excels at serendipity and accuracy. Our results confirm this tradeoff: KNN achieves 99.2% coverage vs. popularity's 0.40%.

Our work differs from prior H&M Kaggle competition entries in two key respects: (1) we emphasize interpretable model comparison over leaderboard optimization, evaluating across four complementary metrics rather than a single ranking metric, and (2) we deploy an interactive application that surfaces recommendations from all three model types, enabling qualitative comparison alongside quantitative evaluation.

4 Evaluation Strategy and Metrics

Metric selection is critical in recommendation systems because different metrics capture fundamentally different qualities. We evaluate on four complementary metrics at K=12:

Hit Rate @ K (HR@12). The fraction of users for whom at least one test-set item appears in the top-K recommendations. This is a coarse binary measure—did the model surface anything relevant? HR rewards breadth: a model that recommends popular items will hit often because popular items appear in many test sets. We include it as the most interpretable metric for non-technical stakeholders.

Normalized Discounted Cumulative Gain @ K (NDCG@12). A position-weighted metric that rewards placing relevant items earlier in the list:

$$\text{DCG@}K = \sum_{i=1}^{K} \frac{\mathbf{1}[\text{item}_i \in \text{test}]}{\log_2(i+1)}, \qquad \text{NDCG@}K = \frac{\text{DCG@}K}{\text{IDCG@}K} \tag{1}$$

We select NDCG as the primary ranking quality metric because in a real shopping interface, users see the first few recommendations and rarely scroll to the end.

Coverage. The fraction of the total catalog (n=2,998) that appears in at least one user's top-K recommendations. Coverage measures diversity at the system level: a model with 0.40% coverage recommends only 12 items total, while 99.2% coverage surfaces nearly the entire catalog across users.

Novelty. The average self-information (in bits) of recommended items:

$$\text{Novelty} = \frac{1}{|U| \cdot K} \sum_{u \in U} \sum_{i \in \text{rec}_u} -\log_2 p(i) \tag{2}$$

where $p(i)$ is item i's popularity (fraction of users who purchased it) in the training set. Higher novelty means the model recommends less popular, more surprising items.

We chose **not** to evaluate precision or recall at K because, in implicit feedback settings, the absence of a purchase does not indicate disinterest—the customer may not have seen the item.

5 Data Processing Pipeline

The pipeline consists of four stages, implemented in modular scripts under `scripts/`:

5.1 Subsampling (`scripts/subsample.py`)

The full H&M dataset (31M transactions) is too large for practical model training on limited compute. We apply three filters:

Table 2: Subsampling filters and rationale.

Filter	Threshold	Rationale
Date	After 2020-08-01	Focus on recent behavior
Customer activity	$\geq$ 15 purchases	Remove inactive users
Article popularity	$\geq$ 50 purchases	Remove niche items

After subsampling: 23,373 customers, 2,998 articles. The interaction matrix has sparsity >99%, typical for fashion e-commerce.

5.2 Feature Engineering (`scripts/build_features.py`)

Temporal train/test split. The last 14 days of the subsampled dataset are held out as the test set. All transactions before this cutoff form the training set. This simulates real-world deployment where models are trained on historical data and evaluated on future behavior. Random splits would leak temporal information.

Interaction matrix. User-item pairs are deduplicated, producing a binary matrix where 1 indicates at least one purchase.

5.3 Image Feature Extraction (`scripts/extract_image_features.py`)

ResNet50 [He et al., 2016] pretrained on ImageNet serves as a feature extractor: classification head removed, global average pooling produces 2,048-dimensional vectors. Standard ImageNet preprocessing is applied (resize 256, center crop 224, normalize with mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Output: `image_embeddings.npy` of shape $(n_{\text{articles}}, 2048)$.

5.4 Metadata Feature Encoding

Article attributes are one-hot encoded across 7 categorical columns (product type, product group, colour group, department, index group, section, garment group), producing a 206-dimensional sparse binary feature vector per article.

6 Hyperparameter Tuning Strategy

Table 3: Hyperparameters for KNN and NCF models.

Model	Hyperparameter	Value	Rationale
KNN	`n_neighbors`	50	Stable user profiles
	Metric	Cosine	Standard for sparse binary vectors
	Image weight	$0.5\times$	Prevent dimensionality dominance
NCF	`embed_dim`	64	Standard for implicit NCF [He et al., 2017]
	`mlp_dims`	[128, 64, 32]	Decreasing-width per NCF paper
	Epochs	10	Sufficient for convergence
	Batch size	1,024	GPU utilization vs. gradient noise
	Learning rate	0.001	Adam [Kingma and Ba, 2015] default
	Negative ratio	4:1	Standard for implicit feedback
	Dropout	0.2	Per MLP layer

The popularity baseline has no hyperparameters—it returns the globally most-purchased items ranked by transaction count.

7 Models Evaluated

7.1 Popularity Baseline (`scripts/train_baseline.py`)

A non-personalized recommender that returns the top-K most purchased items globally. This establishes the performance floor: any personalized model must exceed popularity on at least one metric to justify its complexity.

Rationale. Popularity baselines are surprisingly strong in fashion e-commerce because purchase distributions are heavily skewed—a small number of "hero" products account for a disproportionate share of transactions. The baseline also has a practical deployment advantage: zero latency, zero compute, and it works for any user regardless of purchase history.

7.2 KNN Content-Based (`scripts/train_knn.py`)

A nearest-neighbor recommender that builds user profiles from the mean of purchased article feature vectors, then finds the K nearest articles in feature space.

Rationale. Content-based filtering solves the cold-start problem for items: a new article with metadata can immediately be recommended. It also provides high catalog coverage because recommendations are drawn from the full feature space, not limited to items with interaction history.

Architecture:

1. **Feature matrix.** Each article is represented by its one-hot encoded metadata vector (206 dimensions). Optionally, ResNet50 image embeddings (2,048-dim) are concatenated with $0.5\times$ weighting and L2 normalized.

2. **User profile.** The mean of all purchased article feature vectors, L2 normalized.

3. **Recommendation.** The 50 nearest neighbors to the user profile via brute-force cosine similarity. Top-K items (excluding already-purchased) are returned.

7.3 Neural Collaborative Filtering (`scripts/train_ncf.py`)

A deep learning model that learns non-linear user-item interaction patterns through embedding layers and a multi-layer perceptron [He et al., 2017].

Rationale. NCF combines the collaborative signal of matrix factorization with the expressiveness of deep networks. Adding metadata side features allows the model to leverage both collaborative and content signals.

Architecture:

User and item embedding layers (dimension 64) are concatenated, optionally with metadata features (206-dim) and/or image features (2,048-dim). The concatenated vector passes through an MLP: $[128 \to 64 \to 32]$, each layer applying Linear $\to$ ReLU $\to$ Dropout(0.2). A final Linear(32, 1) $\to$ Sigmoid produces the predicted interaction probability.

Training. Binary cross-entropy loss on implicit feedback. Positive samples are observed purchases (label=1); negative samples are randomly drawn unobserved items (label=0) at a 4:1 ratio. Optimized with Adam [Kingma and Ba, 2015] (lr=0.001).

Model variants: `ncf.pt` (embeddings only), `ncf-meta.pt` (+ 206 metadata features), `ncf-visual.pt` (+ 2,048 image features), `ncf-full.pt` (+ metadata + images).

8 Results

8.1 Quantitative Comparison (K=12)

Popularity wins on hit rate but fails on everything else. The popularity baseline achieves the highest HR@12 (0.0998) because popular items appear in many users' test sets by definition. However, its

Table 4: Recommendation performance across all three models. Best values in **bold**.

Model	HR@12	NDCG@12	Coverage	Novelty (bits)
Popularity (global)	**0.0998**	0.0145	0.0040	8.91
KNN (metadata)	0.0921	**0.0187**	**0.9920**	**11.85**
NCF (+ metadata)	0.0887	0.0178	0.8769	10.68

coverage is catastrophic: 0.40% means only ∼12 items (out of 2,998) are ever recommended. It provides zero personalization.

KNN achieves the best ranking quality. Despite a slightly lower hit rate (0.0921), KNN produces the best NDCG@12 (0.0187), meaning it places correct items higher in the recommendation list. Its 99.2% coverage shows it surfaces nearly the entire catalog across users, and its novelty (11.85 bits) indicates it recommends significantly less popular items than the baseline.

NCF occupies the middle ground. NCF with metadata (0.0178 NDCG, 87.7% coverage) falls between popularity and KNN. The deep model learns useful latent factors but doesn't match KNN's content-based coverage, likely because NCF's collaborative signal is limited by the sparse interaction matrix (>99% sparsity).

8.2 Model Behavior Analysis

The three models serve qualitatively different recommendation strategies:

Popularity returns the same 12 items for every user: bestselling basics like black trousers, white t-shirts, and jersey tops. High purchase probability but zero personalization.

KNN produces highly differentiated recommendations. A customer whose history is concentrated in swimwear receives swimwear, cover-ups, and resort wear. A customer focused on casual lingerie receives comfort-first basics. The content-based approach mirrors the user's existing taste profile.

NCF blends collaborative and content signals. Its recommendations are more diverse than popularity but less taste-specific than KNN, reflecting latent factors learned from the full interaction matrix. NCF occasionally surfaces cross-category items (e.g., recommending a dress to a denim-focused customer) based on collaborative patterns.

9 Error Analysis

We examine 5 specific recommendation failures from the KNN model (our best-performing model on ranking quality). For each case, we identify the recommendation, the user's actual test-set purchase, the root cause, and a concrete mitigation.

9.1 Case 1: Swimwear User Receives Lingerie

- **User profile:** "Beach & Beyond"—13 purchases concentrated in swimwear
- **Recommended:** Casual lingerie bra (article 0659459)
- **Actual test purchase:** Swimwear bikini top
- **Root cause:** One-hot metadata for swimwear and casual lingerie share feature values (similar `index_group`, overlapping `section_name`). The KNN model cannot distinguish "worn at the beach" from "worn at home" using categorical metadata alone.
- **Mitigation:** Add an `occasion` feature or incorporate image features, which could separate swimwear (bright prints) from lingerie (neutral tones, lace).

9.2 Case 2: Heavy Purchaser Gets Redundant Basics

- **User profile:** "The Everyday Essential"—137 purchases in casual lingerie and jersey basics
- **Recommended:** 12 variations of jersey tops and basic tees
- **Actual test purchase:** A knit cardigan (different category)
- **Root cause:** With 137 purchases in a narrow range, the mean-of-purchases profile is dominated by basics. 130 purchases of basics outweigh 7 purchases of knitwear, producing a profile that points directly at more basics. The user's emerging interest in knitwear is invisible.
- **Mitigation:** Time-weighted averaging, giving recent purchases higher weight.

9.3 Case 3: Light Purchaser Gets Generic Recommendations

- **User profile:** "Capsule Wardrobe"—only 5 purchases in light basic jersey
- **Recommended:** Generic popular jersey items (similar to popularity baseline)
- **Actual test purchase:** A structured trouser (different category entirely)
- **Root cause:** With only 5 purchases, the user profile is poorly defined. KNN cannot extrapolate to trousers because nothing in the purchase history signals interest in that category.
- **Mitigation:** For users with <10 purchases, blend KNN with NCF or popularity-based suggestions. NCF's collaborative signal can surface cross-category items.

9.4 Case 4: Activewear User Gets Wrong Sport Category

- **User profile:** "Athleisure"—17 purchases in sport bras and activewear tops
- **Recommended:** Sport shorts and running tights
- **Actual test purchase:** Another sport bra (same narrow category)
- **Root cause:** The one-hot encoding treats all sport items as similar (same `department_name`). The model cannot distinguish "this user buys only sport bras, not sport bottoms."
- **Mitigation:** Increase the weight of `product_type_name` relative to broader categories, or learn feature weights via validation.

9.5 Case 5: Dress User Receives Wrong Style

- **User profile:** "Dress to Impress"—12 purchases, all dresses
- **Recommended:** Casual jersey dresses
- **Actual test purchase:** A structured woven dress
- **Root cause:** Metadata captures "Dress" as product type but does not encode fabric type, structure, or formality. A jersey dress and a tailored woven dress are in the same category but serve different occasions.
- **Mitigation:** Image features could capture visual properties like fabric drape, structure, and pattern.

9.6 Summary of Root Causes

The dominant failure mode is metadata granularity: one-hot encoding captures broad category membership but not within-category distinctions (fabric, occasion, style) that drive purchase decisions.

Table 5: Root causes of recommendation failures.

Root Cause	Cases	Addressable?
Metadata too coarse for subcategories	1, 4, 5	Yes—finer features or image features
Mean-of-purchases amplifies dominant taste	2	Yes—time-weighted profiles
Insufficient purchase history (cold start)	3	Partially—hybrid blending

10 Experiment: Metadata vs. Visual Features

10.1 Hypothesis

If visual features (ResNet50 image embeddings) capture style, color, and silhouette properties that one-hot metadata misses, then augmenting metadata with image features should improve recommendation quality—particularly for within-category distinctions identified in the error analysis (Cases 1, 4, 5).

10.2 Experimental Plan

We train and evaluate four KNN and four NCF configurations on the same temporal train/test split:

Table 6: Feature ablation configurations.

Configuration	Metadata (206-dim)	Images (2048-dim)
Metadata only	✓	—
Visual only	—	✓
Combined	✓	✓(0.5×)
Embeddings only (NCF)	—	—

10.3 Results

Table 7: KNN feature ablation results (K=12).

KNN Configuration	HR@12	NDCG@12	Coverage	Novelty
Metadata only	**0.0921**	**0.0187**	**0.9920**	**11.85**
Visual only	0.0743	0.0134	0.9814	11.62
Combined (meta + visual)	0.0889	0.0172	0.9887	11.71

10.4 Interpretation

Metadata features outperform image features across both architectures. In KNN, metadata-only achieves the best HR@12 (0.0921) and NDCG@12 (0.0187), while visual-only lags behind (0.0743 HR, 0.0134 NDCG). Adding image features to metadata actually *degrades* performance (combined: 0.0889 HR, 0.0172 NDCG). The same pattern holds for NCF: metadata improves over embeddings-only, but adding visual features on top of metadata provides no further gain.

Why do image features underperform? Three factors:

1. **Resolution loss.** H&M product images are low-resolution studio shots (white background, centered garment). ResNet50 features trained on ImageNet capture general visual patterns but may not distinguish fashion-specific attributes like fabric weight, stretch, or drape.

Table 8: NCF feature ablation results (K=12).

NCF Configuration	HR@12	NDCG@12	Coverage	Novelty
Embeddings only	0.0832	0.0156	0.8213	10.31
+ Metadata	**0.0887**	**0.0178**	**0.8769**	**10.68**
+ Visual	0.0841	0.0159	0.8345	10.42
+ Meta + Visual	0.0871	0.0169	0.8592	10.55

2. **Dimensionality imbalance.** Image embeddings are 2,048-dimensional vs. 206 for metadata. Even with $0.5\times$ weighting, image features dominate cosine similarity, diluting the precise categorical signal from metadata.

3. **Metadata already captures what matters.** For H&M's structured catalog, the 7 categorical columns encode the primary axes of purchase decisions. Image features add visual nuance, but that nuance is not the dominant signal in this implicit feedback setting.

10.5 Modeling Decision

This experiment validates deploying KNN with metadata-only features and NCF with metadata as the production models. Image features would require either (a) a fashion-specific visual encoder (not general-purpose ResNet50) or (b) a learned feature weighting scheme. Both represent future work.

11 Interactive Application

The deployed application at `https://one-haut-encoded.github.io` provides a consumer-facing interface for exploring fashion recommendations across all three model types.

Architecture. Static HTML/CSS/JS on GitHub Pages (frontend), FastAPI on Hugging Face Spaces (backend), with a 130 KB precomputed JSON fallback for offline mode. Product images are served from the HF dataset CDN.

Features:

1. **Style Profiles.** 10 demo users with evocative names ("The Everyday Essential," "Denim & Done") spanning 5 to 137 purchases.

2. **Model Toggle.** Users switch between Popularity, KNN, and NCF recommendations to see qualitative differences.

3. **Purchase History.** Each profile displays past purchases with product images.

4. **Curate Your Look.** Users select 3+ items from a curated browse grid and receive real-time KNN-based recommendations, demonstrating the cold-start use case.

5. **Methodology Documentation.** The page includes sections on model methodology, data pipeline, and evaluation results.

The application is inference-only: no training code runs in the deployed environment. Model artifacts are loaded from HF Hub (`DTanzillo/one-haut-encoded`) at startup.

12 Conclusions

We built and evaluated three recommendation architectures on the H&M fashion dataset. The project produced three main findings:

1. Coverage and novelty matter as much as accuracy. The popularity baseline achieves the highest hit rate but recommends only $\sim$12 items out of 2,998—0.40% catalog coverage. KNN

achieves 99.2% coverage with comparable hit rate. In fashion e-commerce, where discovery is the primary value proposition, coverage is arguably more important than marginal hit rate improvements.

2. Structured metadata outperforms visual features. Our ablation study shows that one-hot encoded metadata (206 dimensions) consistently outperforms ResNet50 embeddings (2,048 dimensions) across both KNN and NCF. Adding image features provides no improvement and can slightly degrade performance, likely due to dimensionality imbalance and the generic nature of ImageNet-pretrained features for fashion.

3. Interaction sparsity limits NCF. NCF with metadata achieves respectable performance but does not outperform the simpler KNN model. With >99% interaction matrix sparsity, the collaborative signal is thin. Content-based approaches that leverage rich article metadata are better suited to this data regime.

13 Future Work

1. **Sequential recommendation.** Transformer-based models like SASRec [Kang and McAuley, 2018] or BERT4Rec [Sun et al., 2019] that capture temporal purchase patterns.

2. **Fashion-specific visual encoders.** Replace ResNet50 with CLIP [Radford et al., 2021] or SigLIP embeddings pretrained on fashion image-text pairs.

3. **Learned feature weighting.** Attention mechanisms in NCF to dynamically weight metadata vs. visual features per user.

4. **Richer interaction signals.** Click-through, add-to-cart, and browse data would provide denser implicit feedback.

5. **Time-weighted user profiles.** Recency-weighted profiles to capture evolving taste (addressing Case 2).

6. **Graph neural networks.** GNNs on the user-item bipartite graph for cross-category discovery.

14 Commercial Viability Statement

A fashion recommendation system that achieves high coverage and novelty while maintaining acceptable accuracy has clear commercial value for online retailers. Our system demonstrates three deployment-ready capabilities: (1) multi-model serving enabling A/B testing in production, (2) cold-start handling via the "Curate Your Look" onboarding flow, and (3) offline resilience through precomputed fallback.

Two limitations constrain near-term deployment. First, the H&M dataset is from 2018–2020 and does not reflect current trends; a production system requires continuous retraining. Second, our metrics (HR@12, NDCG@12) measure recommendation accuracy but not business outcomes (revenue, customer lifetime value). A commercially viable system needs online A/B testing to validate that improved recommendations translate to improved business metrics.

15 Ethics Statement

Data provenance. The H&M dataset is publicly available on Kaggle. All customer IDs are anonymized hashes with no personally identifiable information. No scraping, API abuse, or unauthorized data collection was performed.

Recommendation bias. Popularity-based models reinforce existing purchase patterns, creating feedback loops where popular items become more popular. Our KNN model partially addresses this through high coverage (99.2%), but deployed systems should be monitored for diversity degradation.

Filter bubble risk. Content-based models recommend items similar to past purchases, potentially narrowing exposure to new categories. The model toggle in our application provides transparency;

a production system should incorporate diversity-promoting mechanisms (exploration bonuses, serendipity injection).

Environmental impact. All training was performed on single-GPU instances with memory caps. The deployed inference backend runs on CPU.

AI attribution. All code includes AI attribution headers indicating tools used in development (Claude Code, Cursor).

References

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. Neural collaborative filtering. In *Proceedings of the 26th International Conference on World Wide Web*, pages 173–182, 2017.

H&M Group. H&M personalized fashion recommendations. Kaggle Competition, 2022. URL https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendations.

Wang-Cheng Kang and Julian McAuley. Self-attentive sequential recommendation. In *IEEE International Conference on Data Mining (ICDM)*, pages 197–206, 2018.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2015.

Pasquale Lops, Marco de Gemmis, and Giovanni Semeraro. Content-based recommender systems: State of the art and trends. In *Recommender Systems Handbook*, pages 73–105. Springer, 2011.

Julian McAuley, Christopher Targett, Qinfeng Shi, and Anton van den Hengel. Image-based recommendations on styles and substitutes. In *Proceedings of the 38th International ACM SIGIR Conference*, pages 43–52, 2015.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763, 2021.

Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang. BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformers. In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, pages 1441–1450, 2019.